Table 1: Overall performance of different models on four datasets from different domains. Bold indicates the best result for each row and '_' indicates the best result for each LLM. 'random' denotes recommendation based on a random policy. 'pop' denotes recommendation based on items' popularity judged by the number of interactions.

Domain	Metric	random	pop	text-davinci-002			text-davinci-003			gpt-3.5-turbo (ChatGPT)		
				point-wise	pair-wise	list-wise	point-wise	pair-wise	list-wise	point-wise	pair-wise	list-wise
Movie	Compliance Rate	-	-	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	99.98%	100.00%
	NDCG@3	0.4262	0.4761	0.5416	0.5728	0.4990	0.4618	0.5441	0.5564	**0.5912**	0.5827	0.5785
	MRR@3	0.3667	0.4103	0.4824	0.5071	0.4363	0.3998	0.4763	0.4950	**0.5260**	0.5162	0.5167
Book	Compliance Rate	-	-	99.96%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	99.98%	99.80%
	NDCG@3	0.4262	0.4999	0.4889	0.5298	0.4290	0.4585	0.5293	0.4597	0.5075	0.5350	**0.5395**
	MRR@3	0.3667	0.4340	0.4247	0.4646	0.3690	0.3993	0.4665	0.4040	0.4495	0.4774	**0.4800**
Music	Compliance Rate	-	-	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	99.96%	99.80%
	NDCG@3	0.4262	0.4094	0.4623	0.4681	0.4277	0.4732	0.5072	0.4506	0.5201	0.5439	**0.5567**
	MRR@3	0.3667	0.3470	0.4030	0.4082	0.3750	0.4113	0.4448	0.4000	0.4605	0.4830	**0.4950**
News	Compliance Rate	-	-	99.80%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	99.60%
	NDCG@3	0.4262	**0.5444**	0.4483	0.4550	0.5059	0.4880	0.4892	0.4742	0.4826	0.4991	0.5094
	MRR@3	0.3667	**0.4840**	0.3879	0.3936	0.4497	0.4271	0.4294	0.4173	0.4246	0.4354	0.4515

Table 2: Rank of different capabilities of different LLMs-based recommendation models on four datasets from different domains.

Domain	text-davinci-002	text-davinci-003	gpt-3.5-turbo (ChatGPT)
Movie	pair-wise > point-wise ≫ list-wise	list-wise ≈ pair-wise ≫ point-wise	point-wise > pair-wise ≈ list-wise
Book	pair-wise ≫ point-wise ≫ list-wise	pair-wise ≫ list-wise ≈ point-wise	list-wise > pair-wise ≫ point-wise
Music	pair-wise > point-wise ≫ list-wise	pair-wise ≫ point-wise ≫ list-wise	list-wise > pair-wise ≫ point-wise
News	list-wise ≫ pair-wise ≈ point-wise	pair-wise ≈ point-wise > list-wise	list-wise > pair-wise > point-wise

Book: We use the "Books" subset of Amazon[3] dataset that contains user ratings for books.

Music: We use the "CDs & Vinyl" subset of Amazon[3] to conduct experiments on the music domain.

News: We use the MIND-small[4] dataset as the benchmark for news domain.

Following the common practices [12, 21, 41], for the Movie, Book, and Music datasets, we treat ratings above 3 as positive feedbacks (labeled as 1) and otherwise as negative feedbacks (labeled as 0). For the News dataset, we used the original binary feedback labels. In the experiments, we use the titles of the items as description in the prompt.

4.1.2 Evaluation Protocols. After processing, we random sample 500 records on each dataset for evaluation due to the expensive cost. For all experiments, we follow the existing practice [34] and pair one positive item with four randomly sampled negative items as the candidate item list. We set the number of shots as 1 for pair-wise and list-wise, and 2 for point-wise. We report top-K Normalized Discounted Cumulative Gain (NDCG@K) and Mean Reciprocal Rank (MRR@K) with $K = 3$. Furthermore, considering that LLMs may generate some illegal output, that is, results that are not in the candidate set, we introduce the metric "Compliance Rate" to compare the compliance rates between different models, which is defined as the proportion of the number of valid results generated to all test samples, i.e., $\frac{\text{Number of Valid Answers}}{\text{Number of Test Samples}}$.

[3]http://jmcauley.ucsd.edu/data/amazon/
[4]https://msnews.github.io/

4.2 RQ1: Overall Performance

Table 1 shows the results of different LLMs on four different domains. We have the following observation and conclusions:

ChatGPT and GPT3.5s performed much better than the random recommendation in almost all cases. Specifically, all three LLMs achieve significant improvements than the random recommendation policy on four domains, e.g., average improvements with 24.71% on the point-wise task in terms of *NDCG*@3 on the Movie Dataset. Additionally, most answers of LLMs are compliant due to the capability of in-context learning. These results reveal that LLMs have the potential to facilitate recommender systems.

In comparison to the text-davinci-002 and text-davinci-003, ChatGPT exhibits better performance on almost all evaluation metrics for all three ranking capabilities. For instance, ChatGPT outperformed the other LLMs in 22 out of 24 comparisons, including two ranking metrics, three ranking capabilities, and four domain datasets. The only two exceptions were for point-wise ranking in the news domain when compared to text-davinci-003. We attribute ChatGPT's strong performance to its exceptional capacity for language understanding and reasoning, which allows it to effectively comprehend item similarity and make informed ranking decisions.

ChatGPT performs better with list-wise ranking except in the movie domain. On the other hand, text-davinci-002 and text-davinci-003 perform better with pair-wise ranking in most cases. To provide a clear comparison, we have summarized the ranking of the three LLMs with different ranking capabilities in